

docker_class_12(55)

DEPLOYING A NODEJS APP USING DOCKER BY IMPLEMENTING CI/CD

STEP-1: LAUNCH AN INSTANCE WITH T2.LARGE

STEP-2: JENKINS, DOCKER, GIT, TRIVY

STEP-3: PLUGINS --> NODEJS, JDK, SONAR, OWASP DC

STEP-4: CREATE A PIPELINE

STEP-5: OWASP CONFIGURATION

STEP-6: DOCKER IMAGE BUILD & PUSH

STEP-7: CREATE A CONTAINER

when we integrate jenkins to docker we will get permission issue to overcome

`chmod 777 /var/run/docker.sock`

to overcome permission issue

fork the zomoto-app repo

LAUNCH INSTANCE WITH T2.LARGE

SETUP JENKINS

INSTALL GIT

INSTALL TRIVY

`yum install git docker -y systemctl start docker chmod 777 /var/run/docker.sock`

setup jenkins commands

`amazon-linux-extras install java-openjdk11 -y`

`sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhatstable/jenkins.repo`

`sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key`

`yum install jenkins -y`

`systemctl start jenkins`

steps to install trivy:

`wget https://github.com/aquasecurity/trivy/releases/download/v0.18.3/trivy\_0.18.3\_Linux-64bit.tar.gz`

`tar xzvf trivy_0.18.3_Linux-64bit.tar.gz`

`sudo mv trivy /usr/local/bin/`

`Vim .bashrc`

`export PATH=$PATH:/usr/local/bin/`

`source .bashrc`

step 1 and step 2 is done

sonar container will be create

docker run -d --name sonar -p 9000:9000 sonarqube:lts-community

container has created

sonar qube is starting setup password

sonarqube setup is done

as well jenkins also ready

INSTALL THESE FOLLOWING PLUGINS (Eclipse Temurin Installer, SonarQube Scanner, NodeJs Plugin)

Configure Java and Nodejs in Global Tool Configuration

GOTO JDK IN TOOLS

ADD JDK

JDK NAME = jdk17 (jdk-17.0.8.1+1)

CLICK ON INSTALL AUTOMATICALLY CLICK ON INSTALL ADOPTIUM.NET

AND SELECT VERSION (jdk-17.0.8.1+1)

ADD NODEJS

NODE NAME = node16 (NodeJs 16.2.0)

SAVE IT

WE CONFIGURE DONE

NOW SONARQUBE WE HAVE TO INTEGRATE WITH JENKINS

FIRST CREATE USER IN SONARQUBE

user jenkins

Name jenkins

Email: jenkins@gmail.com

Password: 12345678

create

as we created the user this user we have to create token select token1 --> generate--> copy this token

add this in jenkins

in jenkins where to add means goto jenkins --> goto credentials --> goto system --> goto global

credentials --> add credentials--> select secret text--> paste secret-->id sonar-token --> create

integration we have to do

goto-->mange jenkins-->system lo-->add sonar scan-->name:mysonar-->server_url-->add saved credentials as we did up.

next goto manage jenkins-->goto tools -->name:mysonar-->install automatically i.e install from maven central the version is sonarqube scanner 5.0.1.3006 save

as we did sonar is integrated with jenkins earlier how we did webhook same webhook we can add in sonarqube also how to add --> goto administration-->select configuration-->inside webhooks select --> create name:jenkins-->jenkinsurlwith/sonarqube-webhook/-->click on create

Now we will write now pipeline:

Name: DockerProject

```
pipeline {
  agent any

  stages {
    stage('Code') {
      steps {
        git branch: 'main', url: 'https://github.com/devops0014/Zomato-Repo.git'
      }
    }
  }
}
```

goto pipeline script --> select git:git -->give repo url--><https://github.com/Harishjangidi/Zomato-Repo.git>->branch:main-->generate pipeline

git branch: 'main', url: '<https://github.com/devops0014/Zomato-Repo.git>'

check in ci server i.e cd /var/lib/jenkins/workspace/DockerProject/ and do ll

source code came now we will look into clean space

goto dashboard-->create demo job--> basic pipeline hello world i took in the steps i will give cleanWS and next step i will create folder mkdir folder1 if i build the name folder will create and next time if i do build again is it get success or failure it will get failure because already created folder1 normally but this time how many times you did it will get success because in pipeline we given cleanspace

```
pipeline {
  agent any
```

```

stages {
    stage('clean') {
        steps {
            cleanWs()
        }
    }

    stage('Code') {
        steps {
            git branch: 'main', url: 'https://github.com/devops0014/Zomato-Repo.git'
        }
    }
}

```

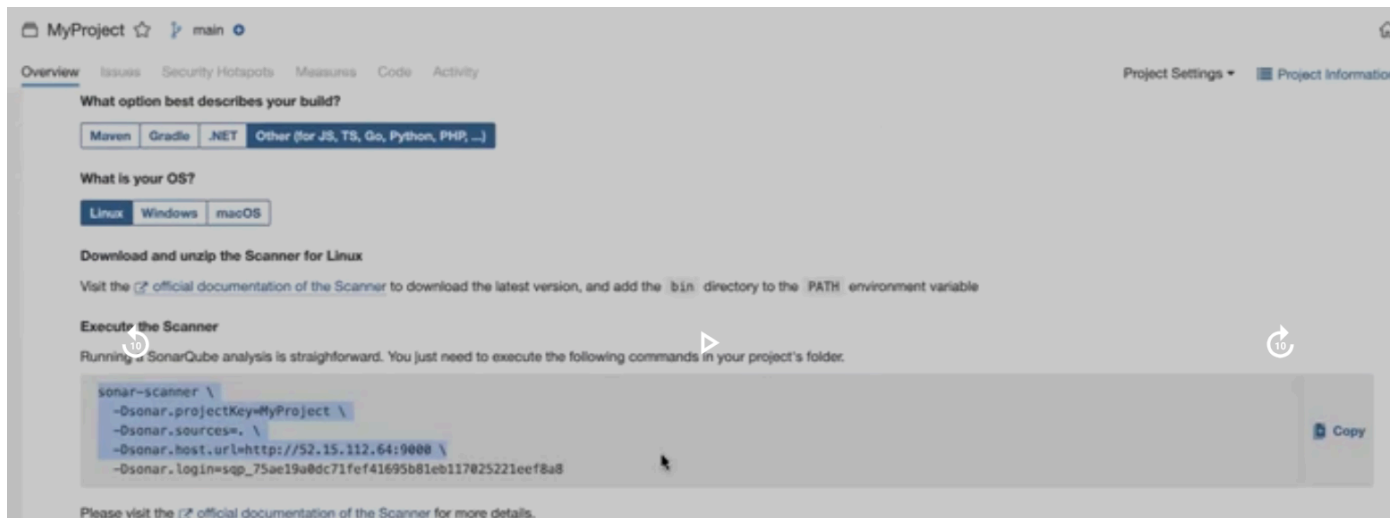
Now next step code quality needs to analysis

what tool we lean i.e sonar qube sonar also provide syntax

click on sonar qube projects click on manually--> select project name: MyProject -->create setup-->--

>select anything i select locally continues -->create token-->select javascript-->select linux os-->

it will provide syntax you can and paste



the above syntax in screenshot we can paste but we have already

already i have this our project just i have updated mysonar in realtime we do like that i have shown

```

stage("Sonarqube Analysis") {
    steps{
        withSonarQubeEnv('mysonar') {
            sh "' $SCANNER_HOME/bin/sonar-scanner -Dsonar.projectName=zomato \
            -Dsonar.projectKey=zomato '"
        }
    }
}

```

below will give full pipeline syntax: we added environment variable and enable the quality gates and with tools we include

```
pipeline {
  agent any

  tools {
    jdk 'jdk17'
    nodejs 'node16'
  }

  environment {
    SCANNER_HOME=tool 'mysonar'
  }

  stages {
    stage('clean') {
      steps {
        cleanWs()
      }
    }

    stage('Code') {
      steps {
        git branch: 'main', url: 'https://github.com/devops0014/Zomato-Repo.git'
      }
    }

    stage('Sonarqube Analysis') {
      steps {
        withSonarQubeEnv('mysonar') {
          sh '''
            $SCANNER_HOME/bin/sonar-scanner \
            -Dsonar.projectName=zomato \
            -Dsonar.projectKey=zomato
          '''
        }
      }
    }

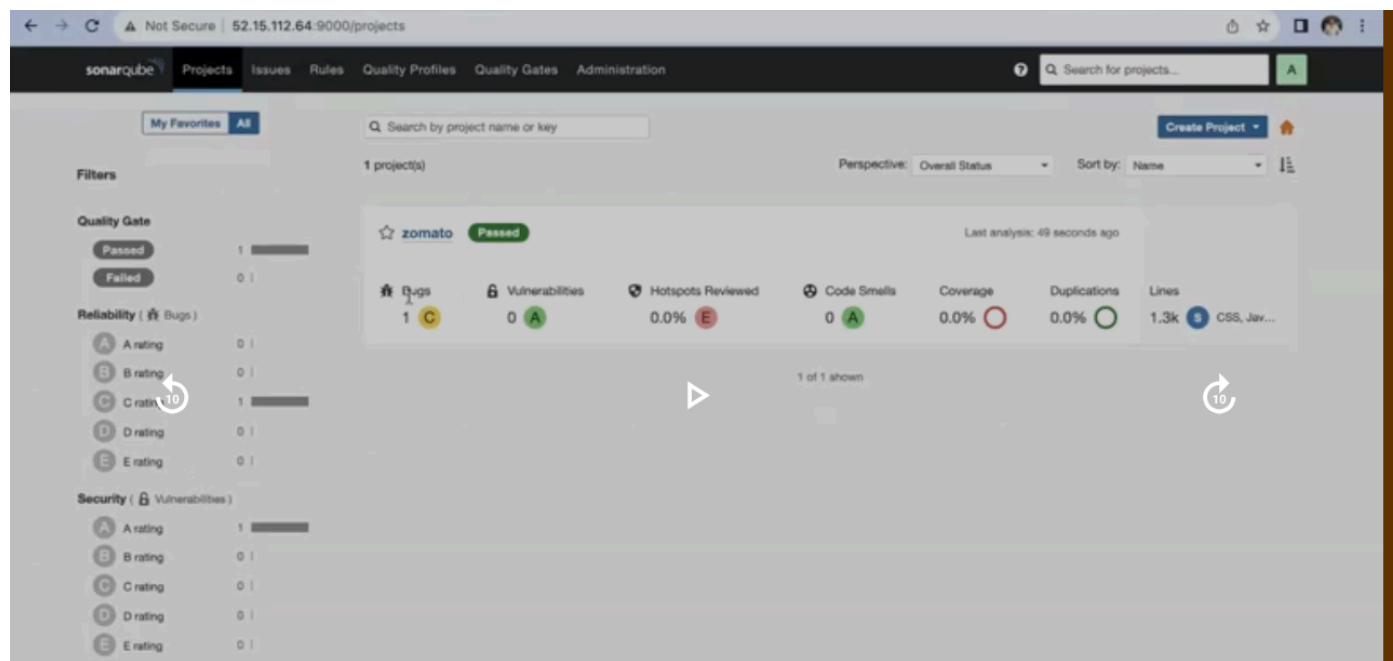
    stage("Quality Gates") {
      steps {
        script {
```

```

        waitForQualityGate abortPipeline: false, credentialsId: 'sonar-token'
    }
}
}
}
}

```

what we face menas generally we did not add tools in pipeline and it is not installed in server i.e in server report.txt file is not genrate so when it is present then only sonarqube analasis will happen just refresh the sonarqube and if you observe we got one bug that we can check



next step is installed the dependency we can check in package.json file also that also how means using owsap
we use npm install

```

pipeline {
    agent any

    tools {
        jdk 'jdk17'
        nodejs 'node16'
    }

    environment {
        SCANNER_HOME = tool 'mysonar'
    }
}

```

```

}

stages {

    stage('clean') {
        steps {
            cleanWs()
        }
    }

    stage('Code') {
        steps {
            git branch: 'main', url: 'https://github.com/devops0014/Zomato-Repo.git'
        }
    }

    stage('Sonarqube Analysis') {
        steps {
            withSonarQubeEnv('mysonar') {
                sh '''
                    $SCANNER_HOME/bin/sonar-scanner \
                    -Dsonar.projectName=zomato \
                    -Dsonar.projectKey=zomato
                '''
            }
        }
    }

    stage('Quality Gates') {
        steps {
            script {
                waitForQualityGate abortPipeline: false, credentialsId: 'sonar-token'
            }
        }
    }

    stage('Install Dependency') {
        steps {
            sh 'npm install'
        }
    }

}
}

```

now it will success will get 2 files as well as node modules

depenncy install is done but how to know is it dependency install correct or not so we can check in owasp plugin download in jenkins -->after that goto jenkins(manage jenkins-->configure tools i.e global tools) dependency check installation-->name whatevr you wish i.e DP-check name --> version is dependency-check 6.5.1(github select) and save

now we will integrate in pipeline now we dont need to remember the syntax-->goto pipeline syntax-->select depany check publisher-->xml report patter i.e **/depany-check-reports.xml--> copy the generate pattern syntax i.e below direct mention in pipeline

dependencyCheckPublisher pattern: '**/dependency-check-report.xml

```
pipeline {
    agent any
    tools {
        jdk 'jdk17'
        nodejs 'node16'
    }
    environment {
        SCANNER_HOME=tool 'mysonar'
    }
    stages {
        stage ("Clean") {
            steps {
                cleanWs()
            }
        }
        stage ("Code") {
            steps {
                git branch: 'main', url: 'https://github.com/devops0014/Zomato-Repo.git'
            }
        }
        stage("Sonarqube Analysis") {
            steps{
                withSonarQubeEnv('mysonar') {
                    sh "' $SCANNER_HOME/bin/sonar-scanner -Dsonar.projectName=zomato \
                    -Dsonar.projectKey=zomato '"
                }
            }
        }
        stage ("Quality Gates") {
```



```

    steps {
      script {
        waitForQualityGate abortPipeline: false, credentialsId: 'sonar-token'
      }
    }
  }
  stage ("Install dependencies") {
    steps {
      sh 'npm install'
    }
  }
  stage ("OWASP") {
    steps {
      dependencyCheck additionalArguments: '--scan ./ --disableYarnAudit --disableNodeAudit',
odcInstallation: 'Dp-Check'
      dependencyCheckPublisher pattern: '**/dependency-check-report.xml'
    }
  }
  stage ("Trivy scan") {
    steps {
      sh "trivy fs . > trivyfs.txt"
    }
  }

  stage ("Build Dockerfile") {
    steps {
      sh 'docker build -t image1 .'
    }
  }
  stage("Docker Build & Push"){
    steps{
      script{
        withDockerRegistry(credentialsId: 'docker-password') {           // this you can get from
pipeline syntax
          sh "docker tag image1 shaikmustafa/loki:mydockerimage"
          sh "docker push shaikmustafa/loki:mydockerimage"
        }
      }
    }
  }
  stage ("Scan image") {
    steps {
      sh 'trivy image shaikmustafa/loki:mydockerimage'

```

```
    }  
  }  
  stage ("Deploy") {  
    steps {  
      sh 'docker run -d --name zomato -p 3000:3000 shaikmustafa/loki:mydockerimage'  
    }  
  }  
}  
}
```

save and continue build

Trivy it will be used like it will give container level it will do security like bugs, vulnerability etc
using Trivy we scan image also

Docker pipeline plugin also needs to install